MIGUEL E. RODRIGUEZ MARTINEZ

> Senior React & Full-Stack Developer

miguelxlerion@gmail.com | +57 313 226 3265 | Bogota, Colombia — Available for Remote / Hybrid

https://mike-portfolio.xlerionultimate.workers.dev/ | www.linkedin.com/in/miguelrodriguez-dataviz

https://github.com/miguelxlerion

PROFILE

Senior Full-Stack Developer with 10+ years of experience specializing in building scalable, high-performance web applications using React.js, Next.js, and TypeScript. Proven track record of architecting robust frontend systems, optimizing load times by up to 35%, and bridging the gap between complex backend APIs and intuitive user interfaces. PMP® certified, ensuring projects are delivered on time and within scope. Currently operating as an independent technical consultant, actively seeking a dedicated, full-time remote role to focus on long-term product development and deep technical challenges.

EXPERIENCE

Consultor sénior independiente de desarrollo full-stack — Sistemas 3D en tiempo real e integración de IA

Feb. 2023 – Present

Xlerion Studio

- I have designed complex front-end applications using React 18 and Next.js, connecting real-time data streams (WebSockets/MQTT) to dynamic and interactive dashboards.
- I have implemented secure authentication flows using JWT and role-based access control (RBAC) for enterprise-level applications.
- I optimised 3D and data-intensive web applications to maintain a stable performance of 60 FPS, reducing data processing time by 40 per cent for end users.
- I managed cloud deployment processes using Docker and Linux environments, ensuring 99 per cent uptime for critical applications.

SKILLS

React.js / Next.js / TypeScript / Tailwind	Avanzado
Node.js / Python / Django / DRF	Avanzado
WebSockets / MQTT / Real-Time Data	Avanzado
PostgreSQL / MySQL / TimescaleDB	Avanzado
LangChain / LlamaIndex / pgvector / RAG	Avanzado
Docker / Nginx / Linux SysAdmin / AWS Cloud Concepts	Avanzado
Three.js / WebGL / Unity / Unreal Engine 5	Avanzado
ZBrush / Blender / 3ds Max / Substance Painter	Avanzado
MaxScript / PBR / Retopology / Pipeline Automation	Avanzado
Celery + Redis / Asyncio	Avanzado

LANGUAGES

Spanish	Native
English	Upper-Intermediate (strong proficiency in technical documentation, code review, and asynchronous communication)

Senior Full-Stack Developer — Governmental IoT Platform

Abr. 2024 – Oct. 2025

Caucasia Town Council (FOVIS)

- Architected and developed a housing management platform from scratch, serving 12+ municipal administrators and managing 1,200+ units.
- Led the creation of a reusable UI component library using React and Tailwind CSS, improving design consistency and responsiveness across the platform.
- Optimized frontend performance, reducing page load times by 35% and improving Core Web Vitals scores.
- Designed real-time KPI dashboards that translated complex backend data into actionable visual insights, reducing administrative decision-making time by 60%.

Web Developer & 3D Integration

2022 – 2023

2INK

- Developed and optimized institutional web environments, ensuring optimal loading times and cross-browser compatibility.
- Collaborated with design teams to translate high-fidelity UI/UX mockups into pixel-perfect, responsive React/JavaScript components.

Frontend Developer & UI Designer

2015 – 2017

USAID - ACDI/VOCA

- Architected the ICOE software frontend, translating complex national-level databases into clear, interactive digital panels for decision-makers.
- Ensured high standards of accessibility, usability, and responsive design for government-level stakeholders.

PROJECTS

Enterprise RAG Assistant

https://github.com/MikeHell84/Rag_X

Use case: reduce search time in unstructured corporate knowledge bases (legal, operations). Achievement: async PDF processing, semantic chunking and hybrid search (vector + lexical) with re-ranking to prevent hallucinations.

Python 3.10+Django/DRFLangChainPostgreSQL + pgvectorCelery + RedisDockerOpenAI/Claude API

IoT Telemetry Resilient Backend

https://github.com/MikeHell84/Ingest_Pipeline

Use case: ingestion of industrial sensor data in environments with intermittent connectivity (mining, logistics). Achievement: fault-tolerant design with local persistence (buffering), guaranteeing 99% uptime under simulated concurrent traffic.

Python AsyncioNode.jsWebSockets/MQTTPostgreSQLLinux SysAdminDocker

CERTIFICATIONS

PMP® Project Management ProfessionalPMI (Project Management Institute) 2020

AI/GenAI SpecializationUC Irvine 2024

Profesional en desarrollo de IA Watson AI Bootcamp 2018

Diploma en Estimación de Proyectos de Software Sergio Arboleda University & MinTIC 2024

Diploma en Gestión Pública Local Cundinamarca Regional Government & FES San José 2025

AI Bootcamp MinTIC & Universidad Sergio Arboleda 2024

Master Power BI Microsoft 2024

Conceptos básicos e integración con ChatGPT Servicios de formación de HPE 2026

Interactive 3D Analytics Dashboard

https://github.com/MikeHell84/xlerion-leaderboard-api

Use case: transform massive tabular databases into real-time gamified visual environments for non-technical users. Achievement: drastic reduction in analysis time while maintaining stable FPS in the browser.

React.jsTailwind CSSThree.js / Unity WebGLWebSocketsDjango REST Framework